SUSTAINABLE TRANSPORTATION COORDINATOR

Carpooling & Bike-Sharing Optimization

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INTRODUCTION

Mobility at the headquarters of the Faculty of Engineering (40th Street, 34th Street and ECCI) presents a critical deficit. The predominant use of private vehicles saturates the available spaces, generating congestion and a negative environmental impact.

Informal initiatives **carpooling** in social networks lack security and institutional monitoring, which limits their mass adoption and effective on the part of the university community.

GOAL

How to optimize them parking spaces and promote sustainable habits through institutional tools?

"Design a system comprehensive coordination that manages the carpooling and loan of bicycles inter-headquarters in a verified Trust Environment."

PROBLEM CONTEXT

~90
MOTORCYCLES ON THE STREET (HEADQUARTERS 40)

55%
ARRIVALS BEFORE 8 AM

The saturation of the basements forces users to parking on public roads, increasing insecurity and road congestion.

PROPOSED SOLUTION

Modular architecture focused on security and optimization resources:

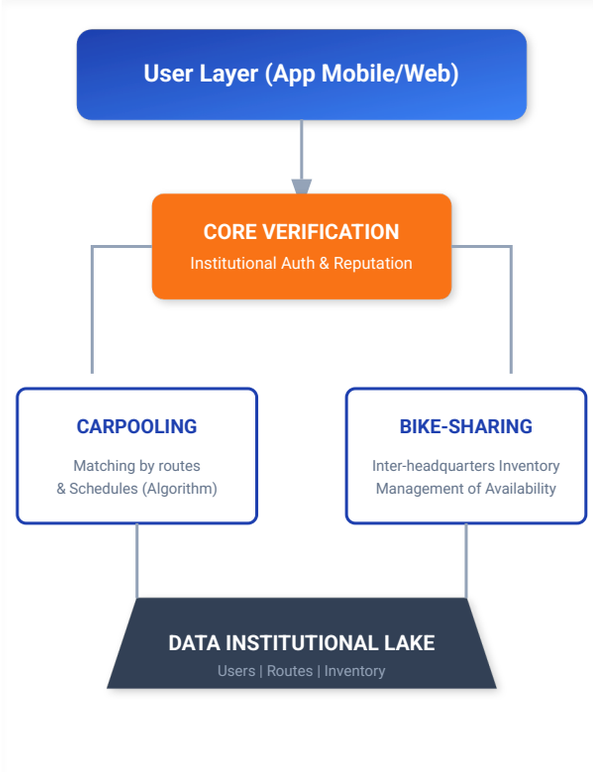


FIG 1. FLOW INTEGRATED SUSTAINABLE SYSTEM

- **Carpooling:** Optimized dynamic matching.
- **Bike-Sharing:** Fleet balancing between headquarters.
- **Verification:** Total user traceability.

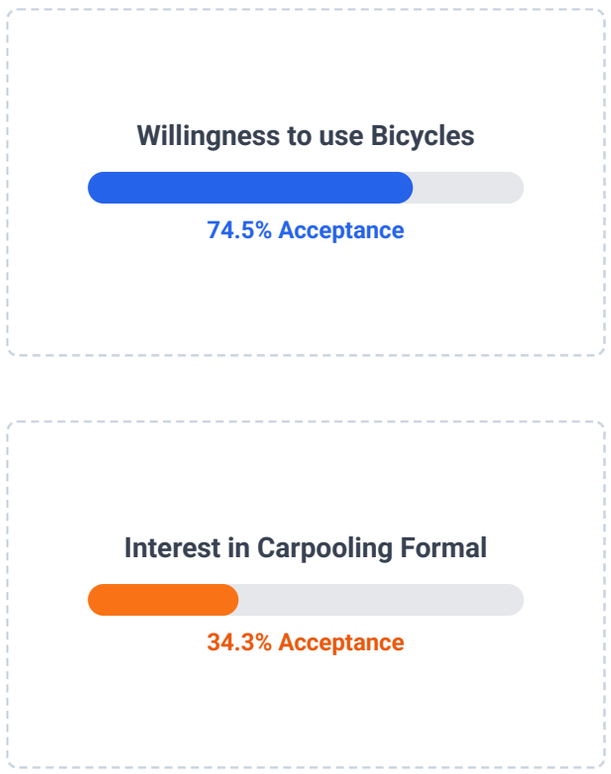
EXPERIMENTS & DATA

Surveys were applied to 51 users in the university community to identify mobility profiles and willingness to change.

ANALYZED CATEGORY	KEY FIND
User profile	98% Students /2% Administrative
Mode of Transport	39.2% Private Vehicle /60.8% Public
Weekly Frequency	47.1% Attend 5 days a week
Interest in Carpooling	**34.3% initial acceptance**

RESULTS

The results project high viability for the bicycle system shared, being the solution preferred by him **74.5%** of the respondents.



CONCLUSIONS

1. The institutional formalization of WhatsApp groups reduces significantly perceived security risks.

2. The bicycle system is the most efficient alternative for the mobility between the headquarters on Calle 40 and Calle 34.

3. It is imperative to integrate physical access control with the database digital to dynamically manage limited parking spaces.

REFERENCES

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